

Biostat D-DCU 200 L - 3:1 microbial vessel (legacy)

sartorius-d-dcu

Field-level manufacturer evidence; remaining values unverified

Each field is classified independently. Direct manufacturer values take precedence over estimates only for the matching model and configuration. Approximate values printed by the manufacturer remain approximate. Unverified fields are editable simulation assumptions, not equipment operating limits.

Parameter	Value in simulator	Evidence status
Minimum working volume	41 L	Source-supported
Maximum working volume	200 L	Source-supported
Nominal vessel / bag size	200 L	Source-supported
Total vessel volume loaded into the form	323 L	Source-supported
Initial working volume loaded into the form	41 L	Derived / estimated
Product-contact material	316L stainless steel / borosilicate glass / EPDM	Source-supported
Height : diameter	3	App assumption
Internal / chamber diameter	0.4395 m	Derived / estimated
Liquid height at maximum working volume	1.32 m	Derived / estimated
Impeller / mixing type	Six-blade Rushton turbine	Source-supported
Impeller count	3	App assumption
Impeller diameter	0.1758 m	Derived / estimated
Power number	5	App assumption
Initial agitation	140 rpm	App assumption
Maximum agitation	570 rpm	Source-supported
Initial aeration	0.5 vvm	App assumption
Maximum aeration	2 vvm	App assumption
Baffled vessel	Yes	App assumption
Maximum pressure	1 bar(g)	App assumption
Default operating pressure	0.2 bar(g)	App assumption
Maximum inlet oxygen fraction	100%	App assumption

Parameter evidence and derivations

Minimum working volume - 41 L

Source-supported | Sartorius Biostat D-DCU brochure - PDF page 16; Culture vessel, 200 L, 3:1, microbial application

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 41 L.

Limitations: Replaces the invalid 10-200 L family envelope with one actual 200 L configuration. Family model sizes are not the turndown range of one vessel.

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[Manufacturer publication](#)

Maximum working volume - 200 L

Source-supported | Sartorius Biostat D-DCU brochure - PDF page 16; Culture vessel, 200 L, 3:1, microbial application

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 200 L.

Limitations: Replaces the invalid 10-200 L family envelope with one actual 200 L configuration. Family model sizes are not the turndown range of one vessel.

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Nominal vessel / bag size - 200 L

Source-supported | Sartorius Biostat D-DCU brochure - PDF page 16; Culture vessel, 200 L, 3:1, microbial application

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 200 L.

Limitations: Replaces the invalid 10-200 L family envelope with one actual 200 L configuration. Family model sizes are not the turndown range of one vessel.

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Total vessel volume loaded into the form - 323 L

Source-supported | Sartorius Biostat D-DCU brochure - PDF page 16; Culture vessel, 200 L, 3:1, microbial application

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 323 L.

Limitations: Replaces the invalid 10-200 L family envelope with one actual 200 L configuration. Family model sizes are not the turndown range of one vessel.

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Initial working volume loaded into the form - 41 L

Derived / estimated | Form-population rule

Rule: initial working volume = min(minimum working volume, maximum working volume)

Selection / calculation: min(41 L, 200 L) = 41 L

Limitations: The preset starts the process at the lower end of its configured working range.

Product-contact material - 316L stainless steel / borosilicate glass / EPDM

Source-supported | Sartorius Biostat D-DCU brochure - PDF page 16; Culture vessel, 200 L, 3:1, microbial application

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 316L stainless steel / borosilicate glass / EPDM.

Limitations: Replaces the invalid 10-200 L family envelope with one actual 200 L configuration. Family model sizes are not the turndown range of one vessel.

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Height : diameter - 3

App assumption | Sartorius Biostat D-DCU brochure - PDF page 16; Culture vessel H:D

Rule: Retained or selected modelling value; not a verified hardware limit.

Selection / calculation: Selected value: 3.

Limitations: Vessel aspect ratio is published; use as liquid H/D remains a model assumption.

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Internal / chamber diameter - 0.4395 m

Derived / estimated | Cylindrical geometry estimate

Rule: $D = \text{cuberoot}[(4 \times V)/(\pi \times H:D)]$

Selection / calculation: $D = \text{cuberoot}[(4 \times 0.2 \text{ m}^3)/(\pi \times 3)] = 0.4394805 \text{ m}$; the form displays 0.4395 m.

Limitations: The estimate treats the liquid volume as a straight-sided cylinder and does not include heads, cones, internals or manufacturer-specific dimensions.

Liquid height at maximum working volume - 1.32 m

Derived / estimated | Cylindrical geometry estimate

Rule: liquid height = estimated internal diameter x H:D

Selection / calculation: $0.4394805 \text{ m} \times 3 = 1.318442 \text{ m}$; the form displays 1.32 m.

Limitations: This is linked to the same ideal-cylinder approximation used for the diameter.

Impeller / mixing type - Six-blade Rushton turbine

Source-supported | Sartorius Biostat D-DCU brochure - PDF page 16; Culture vessel, 200 L, 3:1, microbial application

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: Six-blade Rushton turbine.

Limitations: Replaces the invalid 10-200 L family envelope with one actual 200 L configuration. Family model sizes are not the turndown range of one vessel.

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Impeller count - 3

App assumption | Generic reusable steel microbial profile

Rule: `PROFILE_DEFAULTS["steel-microbial"].impellerCount = 3`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. The count is used by the vessel diagram and represents a generic configuration unless source-supported.

Impeller diameter - 0.1758 m

Derived / estimated | Manufacturer diameter ratio applied to model geometry

Rule: $\text{impeller diameter} = \text{vessel diameter} \times 0.4$

Selection / calculation: $0.4394805 \times 0.4 = 0.1757922 \text{ m}$

Limitations: The ratio is published, but the vessel diameter is estimated. The resulting impeller diameter is therefore derived, not measured.

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Power number - 5

App assumption | Generic reusable steel microbial profile

Rule: `PROFILE_DEFAULTS["steel-microbial"].powerNumber = 5`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. The power number is an empirical generic value used in $P = N_p N^3 D_i^5$. It is not recalculated from blade geometry.

Initial agitation - 140 rpm

App assumption | Large microbial-vessel scaling rule

Rule: $\text{baseRpm} = \text{maxVolume} \geq 500 \text{ L} ? 80 : 140$

Selection / calculation: 200 L is not at least 500 L -> 140 rpm

Limitations: The simulator lowers starting agitation with increasing microbial vessel scale. This is an internal scale heuristic, not a manufacturer limit. This is the initial agitation setting loaded into the form.

Maximum agitation - 570 rpm

Source-supported | Sartorius Biostat D-DCU brochure - PDF page 16; Culture vessel, 200 L, 3:1, microbial application

Rule: Direct manufacturer specification; no reverse calculation.

Selection / calculation: Selected value: 570 rpm.

Limitations: Replaces the invalid 10-200 L family envelope with one actual 200 L configuration. Family model sizes are not the turndown range of one vessel.

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Initial aeration - 0.5 vvm

App assumption | Generic reusable steel microbial profile

Rule: `PROFILE_DEFAULTS["steel-microbial"].baseVvm = 0.5`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This is the initial gas rate loaded into the form.

Maximum aeration - 2 vvm

App assumption | Generic reusable steel microbial profile

Rule: `PROFILE_DEFAULTS["steel-microbial"].maxVvm = 2`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This is the gas-rate ceiling available to the generic DO-control cascade.

Baffled vessel - Yes

App assumption | Generic reusable steel microbial profile

Rule: `PROFILE_DEFAULTS["steel-microbial"].baffled = true`

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This Boolean changes the generic vessel configuration and mixing interpretation.

Maximum pressure - 1 bar(g)

App assumption | Generic reusable steel microbial profile

Rule: PROFILE_DEFAULTS["steel-microbial"].maxPressure = 1

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This is the pressure ceiling used for warnings in the generic equipment model.

Default operating pressure - 0.2 bar(g)

App assumption | Generic reusable steel microbial profile

Rule: PROFILE_DEFAULTS["steel-microbial"].operatingPressure = 0.2

Selection / calculation: No calculation. The vessel inherits this literal value from its generic profile.

Limitations: No model-specific value is stored for this parameter, so the simulator uses its editable reusable steel microbial starting profile. This is the starting headspace pressure loaded into the form.

Maximum inlet oxygen fraction - 100%

App assumption | Fixed simulator capability assumption

Rule: maxOxygenFraction = 100

Selection / calculation: No calculation. Every vessel preset currently permits a 100% oxygen inlet ceiling in the generic gas cascade.

Limitations: This describes the simulator's permitted gas-mixture range, not proof that the selected hardware is certified for pure oxygen service.

Configuration notes and unresolved evidence

This is a family-range preset rather than a single vessel size.

2026-09-11 field-level review: use the evidence beside each parameter; an archived document does not verify the whole preset.

No matching model/configuration-specific primary specification was verified for this field. Legacy values remain editable modelling assumptions, not equipment ratings. Family marketing ranges, external dimensions, sensor ranges, jacket ratings and experimental setpoints are not substituted for vessel specifications.

Local original document identities

Sartorius Biostat D-DCU brochure

sartorius-biostat-d-dcu-brochure-d29d1380d6ff.pdf

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